

Inventory Management System

* Purpose of Inventory Management

The purpose of inventory management is to ensure that the right products are available in the right quantity, at the right time, and at the right cost. It helps businesses control stock efficiently and supports smooth daily operations.

Main Purposes of Inventory Management

1. Ensure Product Availability

To make sure products are always available when customers need them, avoiding stock shortages.

2. Prevent Overstocking

To avoid keeping too much stock, which increases storage costs and risk of damage or expiration.

3. Reduce Operational Costs

To minimize costs related to purchasing, storage, handling, and waste.

4. Improve Stock Accuracy

To maintain accurate records of stock received, sold, returned, or damaged.

5. Support Better Decision-Making

To provide reports and data that help managers make better purchasing and sales decisions.

6. Improve Customer Satisfaction

To fulfill customer orders quickly and correctly by maintaining proper stock levels.

7. Increase Business Efficiency

To automate stock tracking and reduce manual work, saving time and effort.

8. Prevent Loss and Theft

To monitor stock movement and detect missing or suspicious inventory activity.

9. Support Financial Control

To track inventory value and assist with accounting, budgeting, and profit analysis.

10. Help Business Growth

To provide insights for forecasting demand, planning expansion, and managing multiple branches or warehouses.

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* Scope:

-Manage User login & User Permission

-Manage category inventory

-Manage product inventory

-stock movement

-supplier information

-sales transactions for stores or warehouses.

-Report

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- +Core Features:
- Authentication
 - Role Permission
 - Product Management
 - Category Management
 - Inventory/stock management
 - Stock in & stock out tracking
 - Barcode/QR code scanning
 - Supplier management
 - Sales transaction management
 - Invoice & receipt generation
 - Low stock notifications
 - Sales & inventory reports
 - Report

Tech:

Mobile App (Flutter) + Backend (Python or Laravel API) + MySQL

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Inventory Management System Tables & Relationships

++ Table list

1. roles
2. users
3. customers
4. suppliers
5. categories
6. products
7. purchases
8. purchase_details
9. sales
10. sale_details
11. stock_movements
12. warehouses
13. product_transfers

1. roles

Stores user roles.

Column	Type
id	bigint
name	varchar
description	text
disable	boolean
created_at	timestamp
updated_at	timestamp

2. users

Stores login accounts.

Column	Type
id	bigint
role_id	bigint
username	varchar
email	varchar
password	varchar
disable	boolean
created_at	timestamp
updated_at	timestamp

Relationship

- Many users belong to one role
- users.role_id -> roles.id

3. customers

Stores customer information.

Column	Type
id	bigint
user_id	bigint
name	varchar
photo	varchar
phone	varchar
address	text
created_at	timestamp
updated_at	timestamp

Relationship

One customer belongs to one user
customers.user_id -> users.id

4. suppliers

Stores supplier information.

Column	Type
id	bigint
name	varchar
phone	varchar
email	varchar
address	text
created_at	timestamp
updated_at	timestamp

5. categories

Stores product categories.

Column	Type
id	bigint
name	varchar
description	text
created_at	timestamp
updated_at	timestamp

6. products

Stores product details.

Column	Type
id	bigint
category_id	bigint
supplier_id	bigint
name	varchar
barcode	varchar
photo	varchar
cost_price	decimal
sale_price	decimal
qty	integer
unit	varchar
description	text
created_at	timestamp
updated_at	timestamp

Relationship

- Many products belong to one category
- Many products belong to one supplier

products.category_id -> categories.id

products.supplier_id -> suppliers.id

Purchase Module

7. purchases

Stores purchase invoices.

Column	Type
id	bigint
supplier_id	bigint
user_id	bigint
invoice_no	varchar
purchase_date	date
total_amount	decimal
status	varchar
created_at	timestamp
updated_at	timestamp

Relationship

purchases.supplier_id -> suppliers.id

purchases.user_id -> users.id

8. purchase_details

Stores purchased products.

Column	Type
id	bigint
purchase_id	bigint
product_id	bigint
qty	integer
cost_price	decimal
subtotal	decimal

Relationship

purchase_details.purchase_id -> purchases.id
purchase_details.product_id -> products.id

Sales Module

9. sales

Stores sales invoices.

Column	Type
id	bigint
customer_id	bigint
user_id	bigint
invoice_no	varchar
sale_date	date
total_amount	decimal
payment_method	varchar
created_at	timestamp
updated_at	timestamp

Relationship

sales.customer_id -> customers.id
sales.user_id -> users.id

10. sale_details

Stores sold products.

Column	Type
id	bigint
sale_id	bigint
product_id	bigint
qty	integer
sale_price	decimal
subtotal	decimal

Relationship

sale_details.sale_id -> sales.id
sale_details.product_id -> products.id

Inventory Module

11. stock_movements

Tracks stock in/out history.

Column	Type
id	bigint
product_id	bigint
type	enum(in,out,adjustment)
qty	integer
reference_id	bigint
note	text
created_at	timestamp

Relationship

stock_movements.product_id -> products.id

Optional Tables

12. warehouses

Column	Type
id	bigint
name	varchar
location	text

13. product_transfers

Column	Type
id	bigint
from_warehouse_id	bigint
to_warehouse_id	bigint
transfer_date	date

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Main Relationships Diagram

